EXECUTIVE SUMMARY

Credit Risk Analytics & NPA Prediction

FinSight Bank — Analytics Division

Dataset Scale: 2,000,000 Records | Default Subset: loan_status == 1

SECTION 01

Project Objective & Technical Overview

This project establishes an end-to-end credit risk modelling pipeline for FinSight Bank to analyse borrower default patterns and predict **Loss Given Default (LGD%)**. The analysis uses a relational database structure containing **2,000,000 unique records** compiled via sequential left merges on *loan_id*. The primary goals are:

- Identify systemic credit risk drivers and isolate high-risk borrower segments.
- Validate the bank's internal pricing structures against observed default outcomes.
- Test linear and regularised machine learning configurations to improve loan recovery frameworks, satisfying strict OLS and regularisation criteria.

SECTION 02

Data Cleaning & Imputation Strategy

The dataset contained **eight systematically injected data quality anomalies** that required correction without removing any records, preserving the 2,000,000-row architecture.

Anomalies & Rectification

- **Erroneous values** — negative incomes, future approval dates, impossible DTI entries, and out-of-bounds credit scores were isolated, flagged via a binary *dirty_flag*, and replaced using column medians or modes based on variable type.
- **Missing value classification** — high-missing variables (mths_since_last_delinq, mort_acc, emp_length_years, il_util_pct) were analysed. mths_since_last_delinq was classified as *Missing Not at Random (MNAR)* due to reporting thresholds; imputed with column medians or separate category markers.
- **Outlier management** — the top six right-skewed variables were Winsorised at the 1st and 99th percentiles, stabilising linear tracking while preserving all 2,000,000 records.

SECTION 03

Exploratory Data Analysis & Portfolio Insights

Class Imbalance & Credit Baseline

- The portfolio exhibits significant class imbalance; performing accounts substantially outnumber defaulted accounts (class 1). CIBIL scores peak sharply in the 650–750 band, reflecting the bank's historical prime-borrower underwriting bias.

- Overlapping KDE curves for CIBIL scores show that performing and defaulted groups share substantial distributional overlap — confirming that relying solely on static credit scores is insufficient for predictive modelling.

Pricing Monotonicity Validation

- Grouped bar charts confirm that the bank's internal grade system (A through G) is properly structured: interest rates increase step-by-step as credit quality deteriorates, validating the risk-based pricing architecture.

Risk Concentrations & Macroeconomic Shocks

- Speculative categories and small business applications carry the highest default rates; debt consolidation and asset-backed loans show the greatest resilience.
- Timeline analysis (2010–2024) captures a sharp default spike in 2020 attributable to the COVID-19 economic shock. Dual-axis tracking against the RBI repo rate reveals a **12–18 month lag** between rate increases and subsequent borrower default spikes.

SECTION 04

Feature Engineering

Twelve mandatory features were engineered to capture interactive credit risks, each validated via *.describe()* and Pearson correlations against LGD%:

Key Feature Categories

- **Repayment Burden** — emi_to_income_ratio and loan_to_income_ratio show strong positive correlation with loss severity, confirming that high fixed obligations directly amplify default risk.
- **Bureau Behaviour** — delinq_severity_score applies recency weighting to past defaults. This is superior to a simple historical count because a recent missed payment signals immediate financial distress more reliably than an older delinquency.
- **Collateral Protection** — collateral_coverage_ratio shows a strong negative correlation with LGD%. High asset backing ensures that post-default liquidation can substantially recover principal.
- **Distributional Corrections** — log(1+x) transformations on skewed income and loan amounts reduced right-skew to below 0.5, satisfying OLS normality assumptions.

SECTION 05

Statistical Modelling & Regularisation

Data Leakage & Multicollinearity Controls

- Modelling restricted strictly to the defaulted subset (loan_status == 1). All post-default columns (recoveries, collection fees) removed to eliminate data leakage.
- Highly correlated features removed iteratively using a strict VIF threshold (VIF > 10), ensuring stable and independent predictors.

Model Results

- **Baseline OLS** — F-statistic confirmed strong joint significance; Durbin-Watson statistic verified the absence of error autocorrelation.
- **Regularised Models** — Ridge, Lasso, and ElasticNet optimised via GridSearchCV with 5-fold cross-validation. Lasso automatically drove non-essential parameters to zero, producing a streamlined

scoring model.

- **Residual Diagnostics** — 4-panel plot confirmed: (i) linear residual relationships, (ii) log-transformed tail bias correction, (iii) homoscedastic error variance, and (iv) no influential outliers distorting coefficients (Cook's Distance).

SECTION 06

Actionable Credit Risk Recommendations

The following five policy-level recommendations are derived directly from modelling outputs and EDA findings:

#	Recommendation	Key Metric / Threshold
R1	Enforce Hard EMI-to-Income Ceilings	EMI / Income > 45% → Auto-block
R2	Adjust Risk Premiums — Lower Grades	+250 bps for Grade F & G
R3	Tighten Underwriting on Speculative Use Cases	LTV ↓15% + Co-signer required
R4	Deploy Recency-Weighted Bureau Rules	Delinq within 90 days → Line reduction
R5	Mandate Asset Backing for Large Exposures	Collateral ≥ 120% for exposures > ■25 L

Implementation of these measures is expected to materially reduce portfolio LGD% by tightening origination standards, improving bureau-signal responsiveness, and mandating collateral protection on large individual exposures.

This document is intended for internal use by FinSight Bank's Risk, Credit, and Analytics divisions. All figures are derived from the 2,000,000-record modelling dataset.